

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

MINI PROJECT

ACCIDENT ANALYTICS AND VISUALIZATION USING POWER BI

Framework for Data Visual Analytics

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ABSTRACT

This mini project, carried out for the subject Framework for Data Visual Analytics (FDVA), analyses a road accident dataset using Microsoft Power BI. The dataset has 300 accident records and 16 columns for five districts of Tamil Nadu (Chennai, Coimbatore, Madurai, Salem and Tiruchirappalli) from January 2024 to June 2026. It was created in Python with Pandas and NumPy, and it describes each accident by district, location type, road type, weather, time, vehicle type, vehicles involved, fatalities, injuries, severity, speed limit, road condition, traffic level and main cause. The data was checked in Python and had no missing values and no duplicate rows. Twelve visuals were then prepared in Power BI style: bar and column charts, a box and whisker chart, a donut chart, a line chart, a matrix and a scatter plot. The analysis showed that Chennai had the most accidents (81), two-wheelers were involved in the most accidents (122), and failure to signal (44) and overspeeding (43) were the most common causes. Moderate accidents were the largest group (38.3%), and 29.3% of accidents were fatal. Most accidents happened in clear weather and on roads in normal condition. The project shows how simple charts can turn accident records into clear information that can support road safety decisions.

1. Dataset and Data Preparation

The dataset has 300 road accident records and 16 columns. It covers five districts and the period from January 2024 to June 2026. The data was created in Python using Pandas and NumPy (random seed 42) and was checked before the charts were made. Because the data is simulated, the results describe this dataset only and are not real accident statistics.

1.1 Fields in the dataset

Each row is one accident. The main fields are listed below.

- **Accident ID, Date and District:** identifier, date and district of the accident (Chennai, Coimbatore, Madurai, Salem or Tiruchirappalli).
- **Location type, Road type and Road condition:** Urban, Suburban or Rural; National Highway, State Highway, City Road or District Road; and Normal, Wet, Potholes or Under Construction.
- **Weather condition, Accident time and Traffic level:** Clear, Raining or Mist / Fog; a time slot of the day; and Low, Medium or High traffic.
- **Vehicle type and Number of vehicles involved:** Two-wheeler, Car, Auto-rickshaw, Bus, Truck or Lorry, and the number of vehicles in the accident.
- **Fatalities, Injuries and Accident severity:** number of deaths and injuries, and the severity (Fatal, Moderate or Minor).
- **Speed limit and Main cause of accident:** speed limit of the road and the main reason for the accident.

1.2 Checking the data

```
print(df.info())
print(df.isnull().sum())
print(df.duplicated().sum())
print(df.shape)   # (300, 16)
```

The data frame has 300 entries and 16 columns (4 numeric and 12 text). Every column has 300 non-null values, so there are no missing values and no duplicate rows, and no further cleaning was needed.

1.3 Loading the data into Power BI

```
df.to_csv("accident_data.csv", index=False)
```

The data was saved as a CSV file and loaded into Power BI with Get data, and one visual was placed on each of twelve report pages. Counts use Count of Accident ID. Power BI has no built-in box plot or histogram, so a box and whisker visual is used for injuries and a column chart for speed limit.

2. Data Visualization Using Power BI

Twelve visuals were prepared to study accidents by district, vehicle, speed limit, injuries, severity, weather, cause, road condition, traffic level and time. Each chart is explained briefly below.

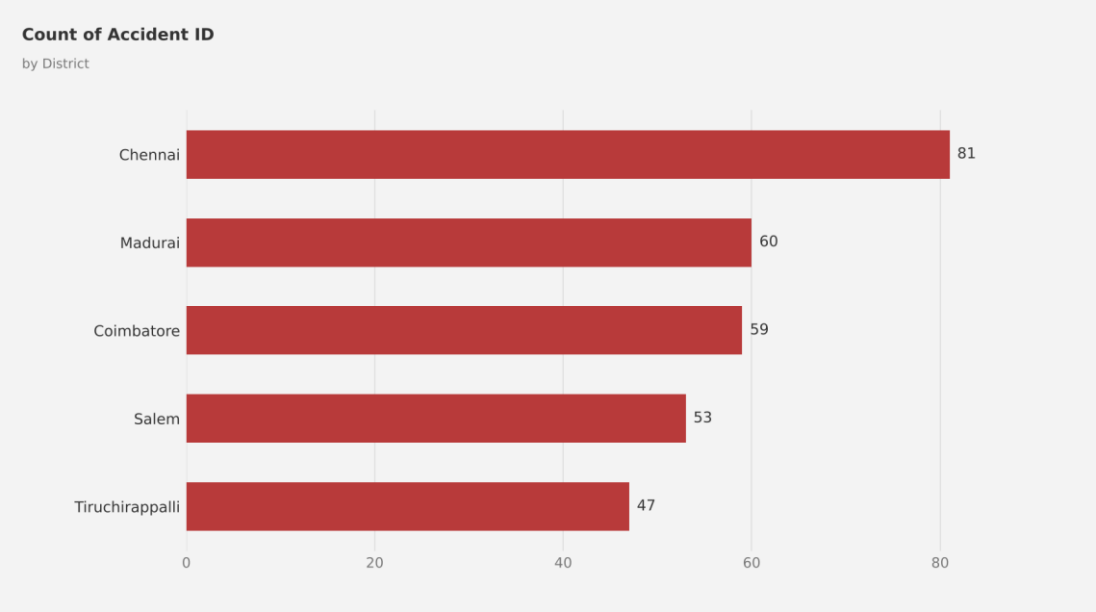


Figure 2.1: Accidents by district (bar chart)

Result: Chennai has the most accidents (81, about 27% of 300), followed by Madurai (60) and Coimbatore (59). Salem has 53 and Tiruchirappalli the fewest (47).

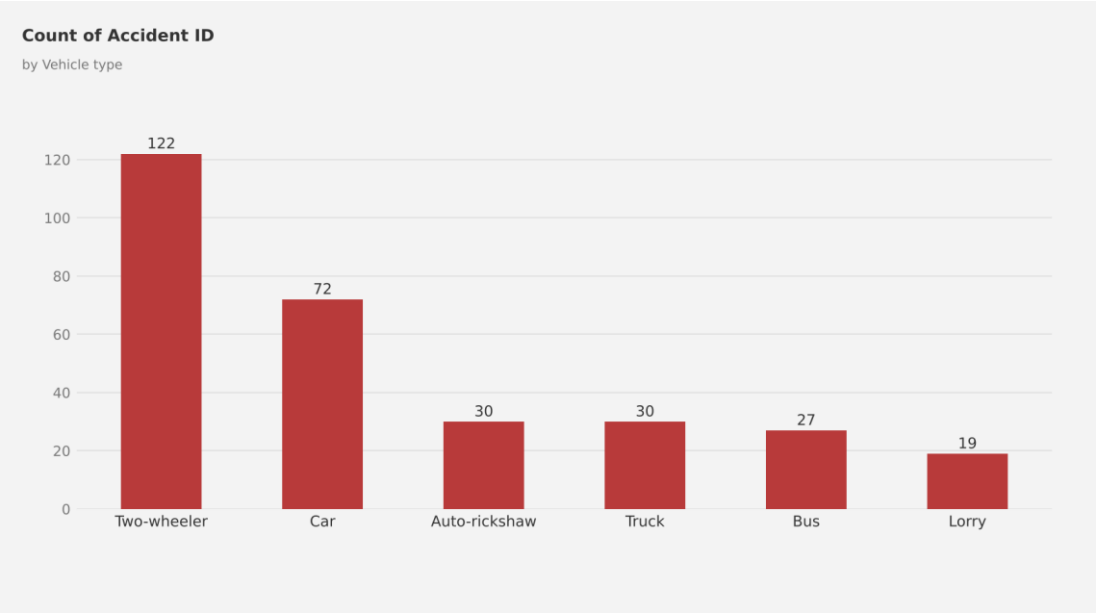


Figure 2.2: Accidents by vehicle type (column chart)

Result: Two-wheelers are involved in the most accidents (122, about 41%), followed by cars (72). Auto-rickshaws and trucks have 30 each, buses 27 and lorries 19.

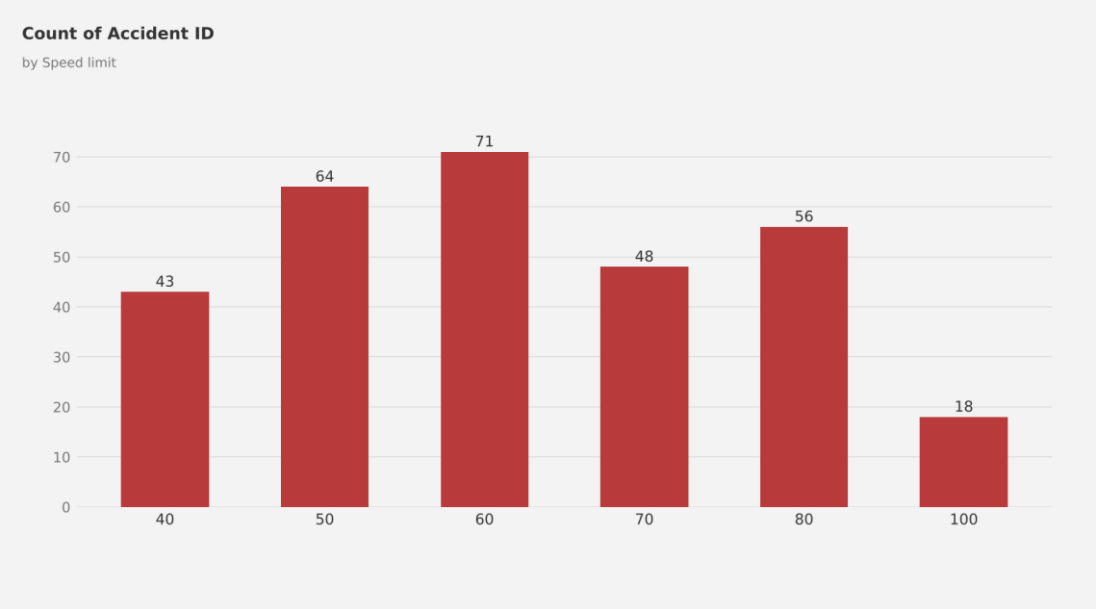


Figure 2.3: Accidents by speed limit (column chart)

Result: Most accidents happen on roads with a speed limit of 60 (71 accidents), then 50 (64) and 80 (56). The speed limit of 100 has the fewest (18).

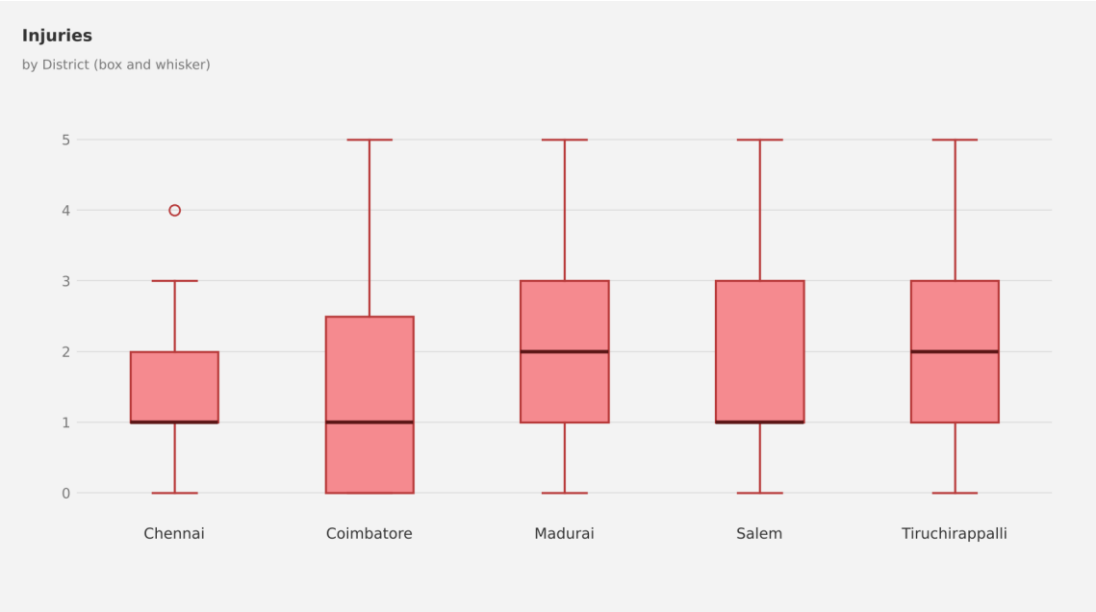


Figure 2.4: Injuries by district (box and whisker chart)

Result: The median number of injuries is 1 in Chennai, Coimbatore and Salem and 2 in Madurai and Tiruchirappalli. Chennai has the narrowest spread (0 to 3, with one outlier at 4); the other four districts reach 5 injuries.

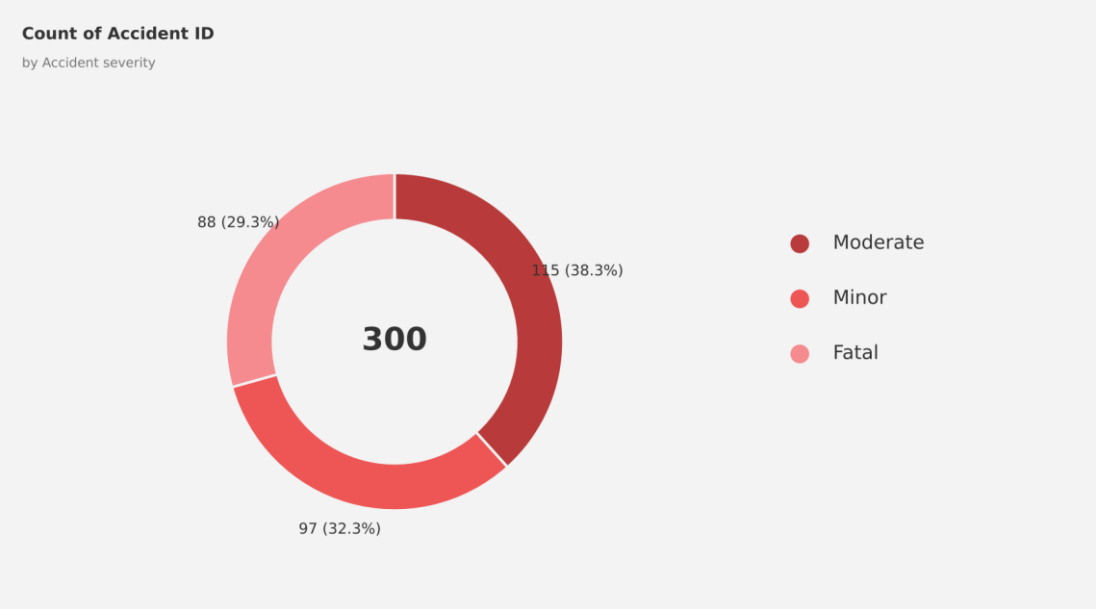


Figure 2.5: Accident severity (donut chart)

Result: Moderate accidents are the largest group (115, 38.3%), followed by minor (97, 32.3%) and fatal (88, 29.3%).

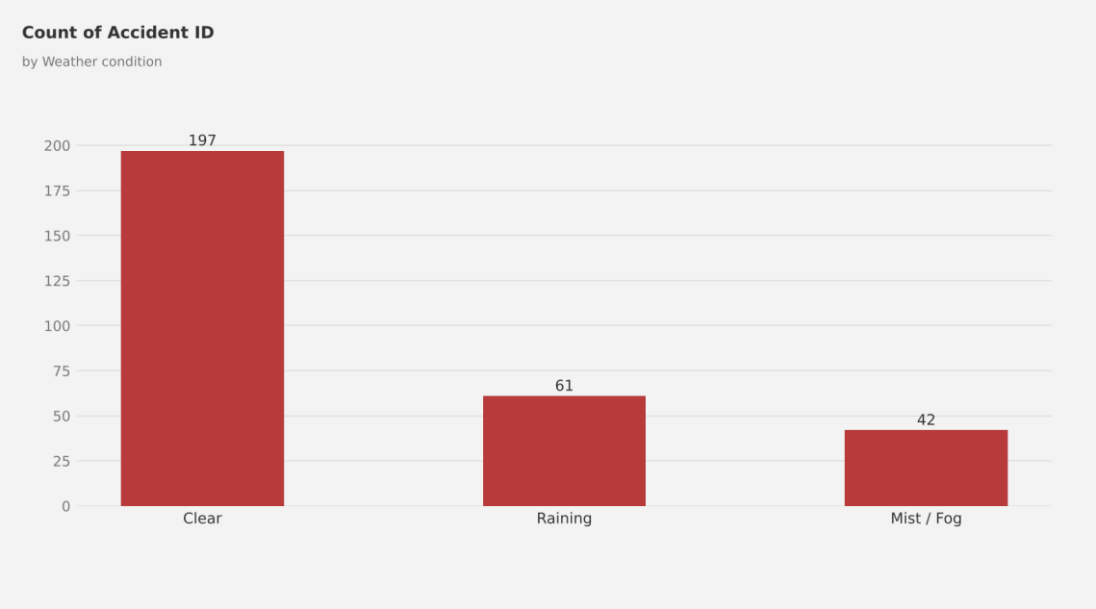


Figure 2.6: Accidents by weather condition (column chart)

Result: Most accidents happen in clear weather (197, about 66%). Raining has 61 and mist / fog has 42.

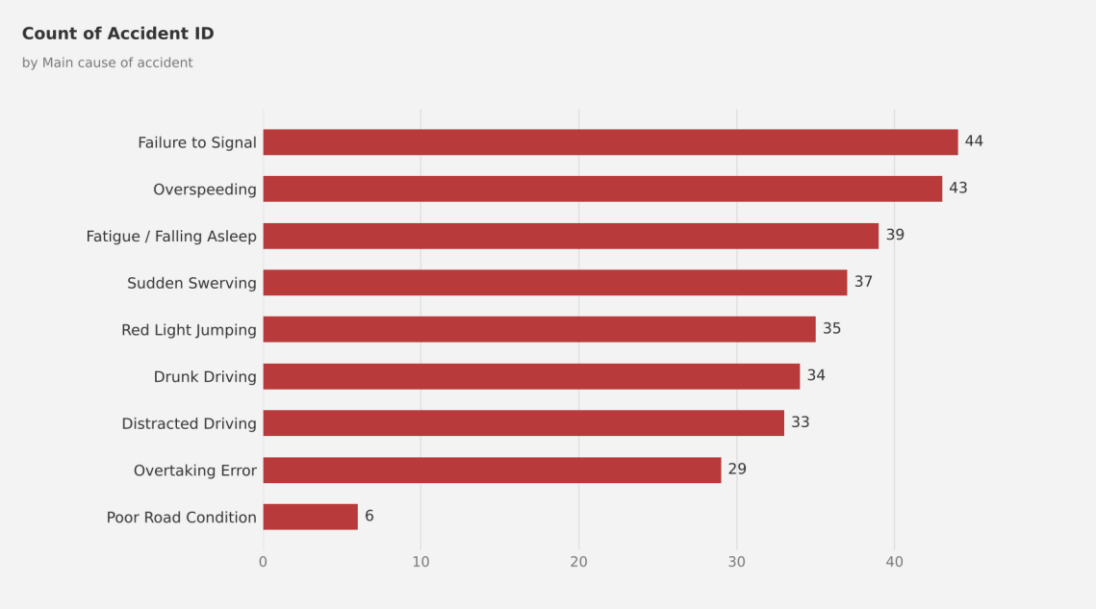


Figure 2.7: Main causes of accidents (bar chart)

Result: Failure to signal (44) and overspeeding (43) are the top causes, followed by fatigue / falling asleep (39) and sudden swerving (37). Poor road condition is the lowest (6). The eight main causes are close to each other (29 to 44).

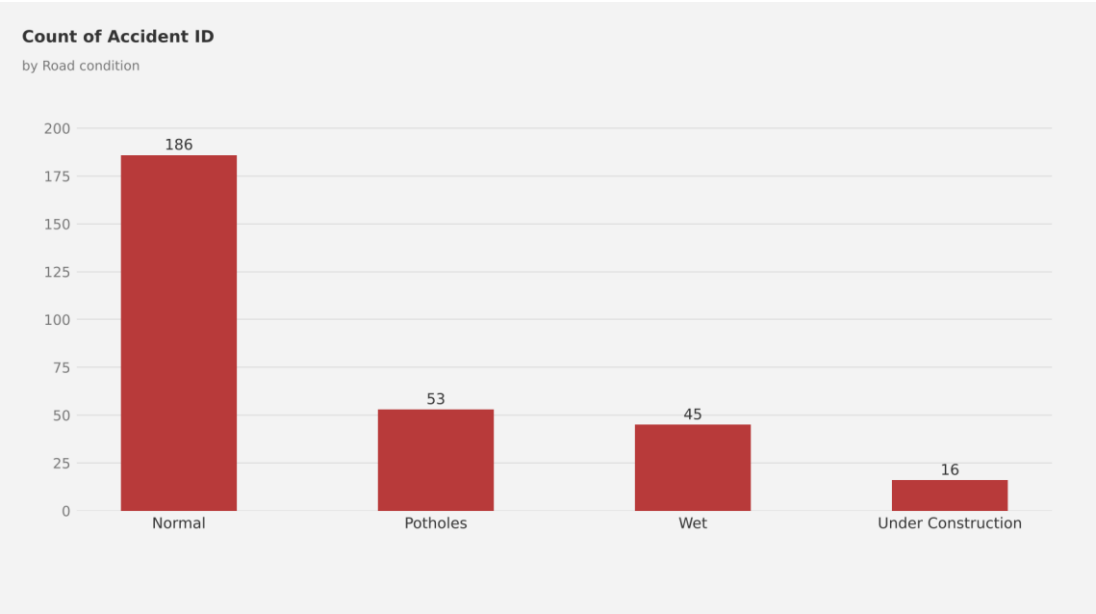


Figure 2.8: Accidents by road condition (column chart)

Result: Most accidents (186, about 62%) happen on roads in normal condition. Potholes (53), wet roads (45) and roads under construction (16) follow.

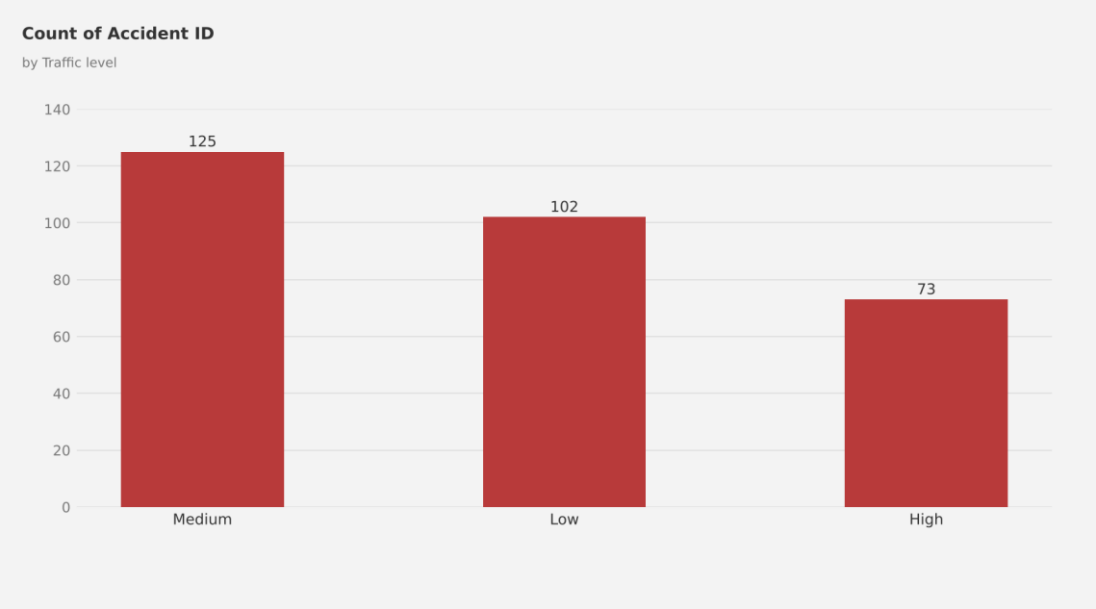


Figure 2.9: Accidents by traffic level (column chart)

Result: Medium traffic has the most accidents (125), followed by low (102) and high traffic (73).

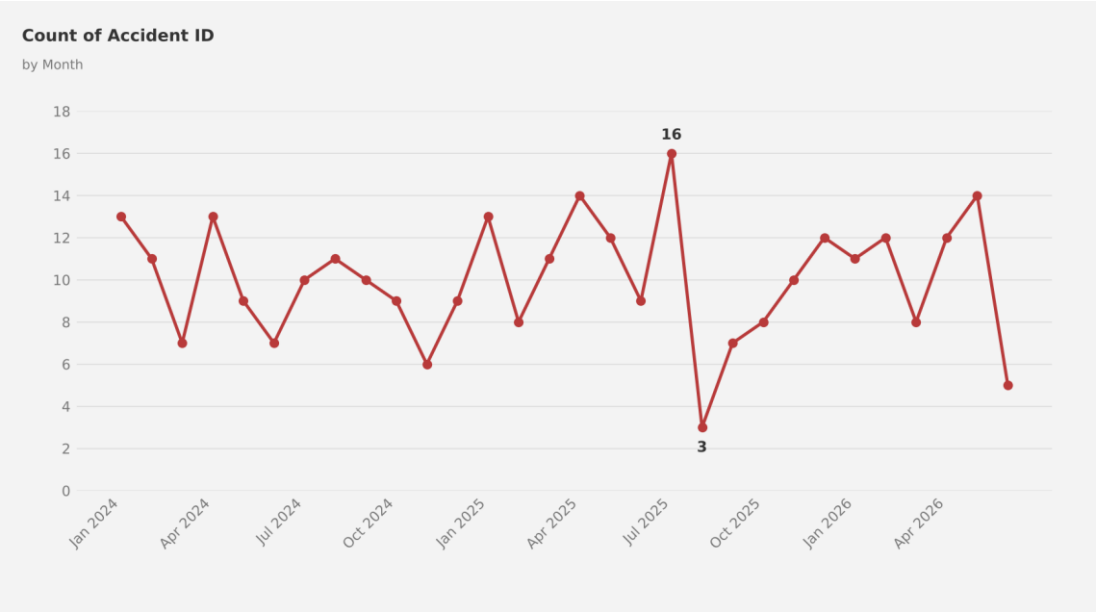


Figure 2.10: Monthly accident trend (line chart)

Result: Monthly accidents vary between 3 and 16 with no steady rise or fall. The highest month is July 2025 (16) and the lowest is August 2025 (3). June 2026 has 5 accidents.

Count of Accident ID

by District and Weather condition (matrix)

District	Clear	Mist / Fog	Raining	Total
Chennai	55	10	16	**81**
Coimbatore	43	7	9	**59**
Madurai	43	6	11	**60**
Salem	33	10	10	**53**
Tiruchirappalli	23	9	15	**47**
Total	**197**	**42**	**61**	**300**

Figure 2.11: Accidents by district and weather condition (matrix)

Result: Clear weather has the most accidents in every district, with Chennai highest (55). The lowest value is Madurai in mist / fog (6). The matrix totals match the district counts and add up to 300.

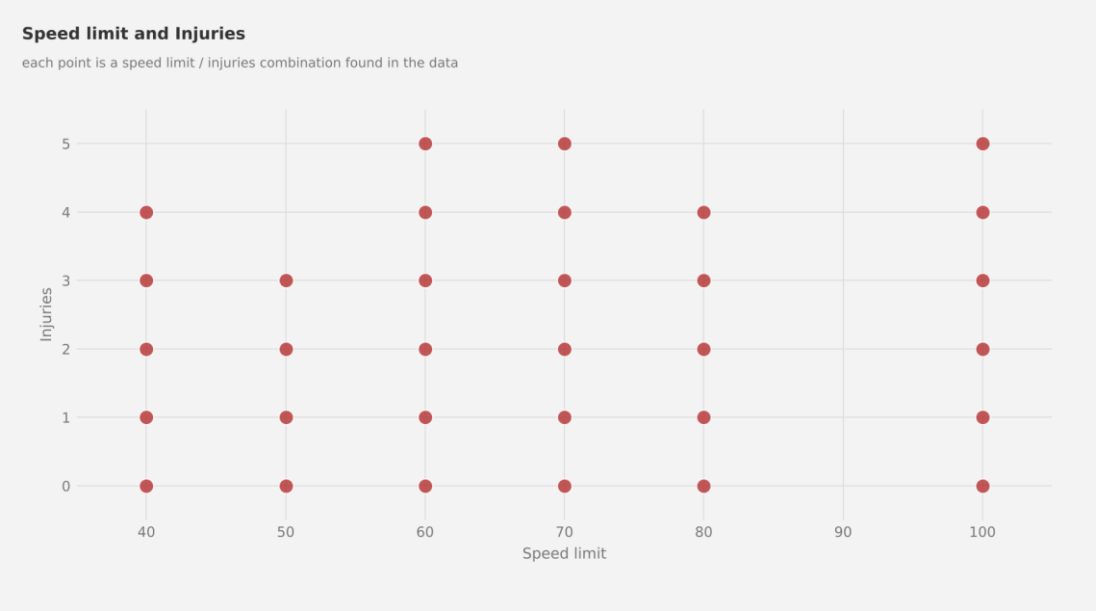


Figure 2.12: Speed limit vs injuries (scatter plot)

Result: Each point is a speed limit and injuries combination found in the data. Injuries range from 0 to 5, and accidents with 5 injuries occur at speed limits 60, 70 and 100. Injuries do not rise steadily with the speed limit.

Overall, accidents are concentrated in Chennai, among two-wheelers, in clear weather and on roads in normal condition, while the causes are spread fairly evenly.

3. Key Findings

Comparing the charts together gives the following points.

- **Where:** Chennai has the highest share of accidents (27%), and Tiruchirappalli the lowest (about 16%).
- **Who:** two-wheelers and cars together are involved in about 65% of accidents.
- **Severity:** nearly one accident in three is fatal (29.3%), and only about a third are minor.
- **Causes:** human behaviour (failure to signal, overspeeding, fatigue, swerving, red light jumping and drunk driving) accounts for most accidents; poor road condition is the main cause in only 6.
- **Conditions:** most accidents are in clear weather and normal road condition, so these counts show where accidents were recorded and not which conditions are riskier.
- **Time:** monthly counts change a lot (3 to 16) without a clear trend.
- **Speed:** the number of injuries does not rise steadily with the speed limit in this data.

4. Conclusion

This mini project used Power BI-style visuals to analyse 300 road accident records from five districts. The charts show that Chennai, two-wheelers, clear weather and normal road conditions have the most recorded accidents, that failure to signal and overspeeding are the most common causes, and that almost 30% of accidents are fatal. These results can help road safety teams focus on two-wheeler safety, signal discipline and speed control in the busiest districts. Because the data is simulated, the same steps should be repeated on real accident records before drawing real-world conclusions.